

Regional Quantitative Problem-Solving Circle

Module 2: Discrete Probability & Combinatorics (Weeks 3–4)

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1 Axiomatic & Theoretical Foundation

Combinatorial counting often becomes computationally intractable when events exhibit complex, overlapping dependencies. We resolve this by mapping sample spaces to real numbers via random variables and analyzing their expected behavior as linear combinations.

Definition 1 (Discrete Probability Space & Random Variable). *A discrete probability space is a tuple $(\Omega, \mathcal{F}, \mathbb{P})$ where Ω is a countable sample space and $\mathbb{P}$ is a probability measure. A random variable is a measurable function $X : \Omega \rightarrow \mathbb{R}$.*

Definition 2 (Indicator Random Variable). *Let $A \subseteq \Omega$ be an event. The indicator random variable I_A is defined as:*

$$I_A(\omega) = \begin{cases} 1 & \text{if } \omega \in A \\ 0 & \text{if } \omega \notin A \end{cases}$$

The expectation simplifies precisely to the probability of the event: $\mathbb{E}[I_A] = 1 \cdot \mathbb{P}(A) + 0 \cdot \mathbb{P}(A^c) = \mathbb{P}(A)$.

2 The Core Invariant / State Function

The most critical invariant of the expectation operator is its absolute linearity, regardless of the underlying joint distribution or covariance between variables.

Theorem 3 (Linearity of Expectation (LoE)). *For any finite sequence of random variables $X_1, X_2, \dots, X_n$ defined on the same probability space, and any constants $c_1, c_2, \dots, c_n \in \mathbb{R}$:*

$$\mathbb{E} \left[\sum_{i=1}^n c_i X_i \right] = \sum_{i=1}^n c_i \mathbb{E}[X_i]$$

*This holds **with absolutely no requirement of independence** between the variables X_i .*

3 Lemma & Canonical Proof

To demonstrate how LoE circumvents dependency structures, we will evaluate the expected number of fixed points in a random permutation. Calculating the exact probability distribution of fixed points requires the Principle of Inclusion-Exclusion and derangement formulas, but calculating the expectation requires merely two lines of structural reasoning.

Lemma 4 (Expected Fixed Points of Uniform Permutations). *Let S_n be the symmetric group of all permutations of the set $\{1, 2, \dots, n\}$. If a permutation $\pi \in S_n$ is selected uniformly at random, the expected number of fixed points (elements where $\pi(k) = k$) is exactly 1, independent of n .*

Proof. Let the random variable X denote the total number of fixed points in the randomly chosen permutation π . We wish to compute $\mathbb{E}[X]$.

We decompose X into a sum of local indicator variables. Define I_k for each $k \in \{1, 2, \dots, n\}$ as:

$$I_k = \begin{cases} 1 & \text{if } \pi(k) = k \\ 0 & \text{if } \pi(k) \neq k \end{cases}$$

By definition, the total number of fixed points is the sum of these indicators:

$$X = \sum_{k=1}^n I_k$$

We now evaluate the expectation of a single indicator, $\mathbb{E}[I_k]$. The probability that a specific element k is mapped to itself is the number of permutations fixing k divided by the total number of permutations. There are $(n-1)!$ permutations that fix k (as the remaining $n-1$ elements can be permuted arbitrarily). Therefore:

$$\mathbb{E}[I_k] = \mathbb{P}(\pi(k) = k) = \frac{(n-1)!}{n!} = \frac{1}{n}$$

Notice that the indicators I_k are highly dependent (e.g., if $n-1$ elements are fixed, the n -th element is deterministically fixed). However, by the Linearity of Expectation, this dependence is entirely irrelevant to the sum of the expectations:

$$\mathbb{E}[X] = \mathbb{E}\left[\sum_{k=1}^n I_k\right] = \sum_{k=1}^n \mathbb{E}[I_k] = \sum_{k=1}^n \frac{1}{n} = n \times \frac{1}{n} = 1$$

Thus, the expected number of fixed points is 1, regardless of the size of the set. □

4 Seminar Heuristics & Socratic Framework

During the whiteboard defense phase, push students on the following diagnostic cues and common pitfalls:

- **Diagnostic Cue - "When to apply?":** Whenever a problem asks for an "expected total," "expected number of pairs," or "average count" in a highly chaotic or combinatorial system, immediately mandate the definition of local indicator variables.
- **Pitfall - The Independence Fallacy:** Students consistently hesitate to add expectations if the events influence one another. *Socratic Prompt:* "Are you multiplying probabilities or adding expectations? Show me the step in the definition of the expectation summation where independence is required. (Hint: It doesn't exist)."
- **Pitfall - Overcomplicating the State Space:** Students will try to find the probability of exactly k events occurring. *Socratic Prompt:* "Do we need the full probability mass function $P(X = k)$ to find the center of mass? Separate the variables."